

Batuhan DURUKAL

(+90)538 0659295 · **batuhan.durukal@gmail.com** · Istanbul, Turkey

Profile

Versatile and innovative engineer with over seven years of experience in the automotive industry, now transitioning expertise into scalable machine learning (ML) systems and artificial intelligence (AI). Proficient in Python, SQL, and modern ML frameworks, with a strong background in data-driven methodologies, software development, and large language models (LLMs). Passionate about designing and implementing cutting-edge ML solutions, leading technical teams, and collaborating across departments to deliver impactful, scalable applications.

Work Experience

AVL Türkiye Jan 2023 - Present
Team Leader - Validation & Data Analytics
- Leadership (up to 8 engineers).
- Business Strategy.
- Innovation Strategy.
- Customer Management.
- RFQs*(Request-for-quotation)* & Negotiations.
- AI applications:
  - LLM-based benchmark reporting.
  - AI-Assistant with RAG via LLM.
  - Evaluation of LLM workflows using LLM-as-a-Judge.
  - Tag categories of customer experience comments via LLM.
  - User-Centric Assessment ML Modeling.
  - Naturalistic Driver Behavior Analysis via Explainable AI (XAI).

AVL Türkiye Mar 2022 - Jan 2023
Senior Data Analyst
- Led the "Validation & Data Analytics" chapter on Department
- Developed a User Experience Rating ML Model for Autonomous Vehicle Lane Change Maneuvers.
  - Investigated safety and comfort performance among determined KPIs using regression algorithms (e.g., XGBoost, Random Forest, Gradient Boosting) on training and testing datasets.
  - Tuned hyperparameters via 5-fold cross-validation grid-search
  - Achieving high performance (measured by R^2) with 22 selected features
  - demonstrating scalable ML solutions for autonomous systems validation.
- Developed the "Maneuver Detection Framework" in Python, integrating ML techniques for real-time analytics.
- Automated workflows with Python tools, enhancing efficiency in KPI tracking and feature validation.
- Test case generation, execution, and reporting for ADAS/AD features.

AVL Türkiye Dec 2018 - Mar 2022
Data Analyst
- Built and optimized ML models for automotive feature performance targets and KPI performance analysis.
- Contributed to the software development lifecycle (SDLC) adhering to Agile practices.
- Developed data pipelines and conducted large-scale data analysis to support feature validation.
- Root cause analysis on anomalies.

Bertrandt Sep 2018 - Dec 2018
Design Engineer
- Designed 3D models using CATIA, collaborating with teams to meet project specifications.

OTOKAR Automotive and Defence Industry Jul 2017 - Apr 2018
Project Engineer
- Responsible for planning, feasibility, and release of Commercial Vehicle Powertrain element designs.

Education

Istanbul Technical University, Istanbul 2023 - 2026
MSc - Mechatronics, Robotics, and Automation Engineering

Istanbul Technical University, Istanbul, Turkey 2013 - 2018
BSc - Mechanical Engineering
BSc - Manufacturing Engineering (Double Major)

Publications

- **How Can Machine Learning Models Be Used for Subjective Assessment of Safety and Comfort: Application on Free Lane Change Maneuver**, *2022 Joint 12th International Conference on Soft Computing and Intelligent Systems and 23rd International Symposium on Advanced Intelligent Systems (SCIS&ISIS)*, Japan, 2022.
- **A Digital Twin Study: Particle Swarm Optimization of ACC Controller for Follow Acceleration Maneuver**, *2022 IEEE 21st international Conference on Sciences and Techniques of Automatic Control and Computer Engineering (STA)*, Tunisia, 2022.
- **Safety Assessment of Partially Automated Lane Change System**, *2021 IEEE 17th International Conference on Intelligent Computer Communication and Processing (ICCP)*, Romania, 2021.

Skills

Programming: Python, SQL, MATLAB
Machine Learning Frameworks: Scikit-learn, TensorFlow, PyTorch, LangChain
Tools: Git, Docker, Poetry, Pydantic, Jira, Confluence, Azure Cloud, OpenAI
SDLC Practices: Agile Methodologies, Code Reviews, Testing, Deployment
Languages: Turkish (Native), English (Fluent), German (Intermediate)

Personal Interests

Board Games, Fitness, Football, Traveling,